

Steered affect moves a conversational partner’s activations: the estimator, three pre-registered ablations, and which controls bite

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Abstract

When one language-model agent is steered into an emotion and talks to another, a linear probe’s projection of the second agent’s reply activations moves with the dose, for five of six emotions on Qwen3.6-27B and on Llama-3-8B. We ask what carries it, and find that the standard linear tools mislead in ways their usual checks do not catch. The logistic-regression direction most work uses to read and write such states reproduces across disjoint halves at only 0.57 (27B, $n = 2000$ per half) and 0.29 (8B), while difference-of-means reaches 0.97 and 0.94 on the same splits, and a fixed per-sample penalty does not close the gap. In the $n < d$ regime probe studies work in, every logistic fit is a perfect separator: its direction is the penalty’s tie-break, drifting from the mean difference as n grows (cosine 0.97 to 0.63); projected onto 100 principal components, where $n \gg d'$, the fits stop separating and the direction reproduces at 0.90. The cost is largely geometric: the two estimators’ held-out projections correlate at 0.90–0.96. With the reproducible direction, three pre-registered ablations test the linear account of the channel. Removing the emotion direction at every readable layer leaves a median 91% of the transfer against a random-direction control (85% against baseline); a rank-5 affect subspace fails its own instrument check; a rank-5 subspace fitted by the same estimator on permuted labels blocks three emotions as strongly as the affect subspace does, while a random 5-frame of half its footprint and a permuted-label rank-1 direction are inert. The emotion direction’s blocks of afraid (25%) and sad (29%) survive the rank-1 permuted control; across models only sad’s replicates. Whether rank or footprint makes the rank-5 control bite is untested, and the receiver’s “other-speaker” projection rises with dose as much as its own, so the probe does not separate the receiver’s state from its model of the sender’s.

1 Introduction

Systems of several language-model agents built from one model family are now common, and the agents talk to each other [3, 8]. If one of them is in an emotional state, does anything of that state reach the others? A transfer that changes the receiver’s internal representation would be invisible in transcripts and visible only in activations, which makes it a question for representation-level tools: steering vectors to induce a state in the sender [7, 10, 13, 14], probes to read it in the receiver, and ablations to test what carries it.

Those tools rest on one object, a direction in activation space. This paper’s single idea is that the direction has to be trusted twice, once to read and once to intervene, and that both uses fail

*Revised draft of 2026-09-06 after three adversarial reviews, a mentor review, a citation audit and a statistical-consistency audit (all in the repository). Every number is registered in `docs/review/claims.json` under the identifier printed after it in monospace; the registry is machine-checked against the result files. Figures and tables are generated from those files by `src/rev3/paper_figures.py` and `src/rev3/paper_revision_stats.py`.

in ways the usual checks miss. Read with a logistic-regression coefficient row, the state looks like a moving target: the direction does not reproduce across independent fits, and the reason is not the sample size or the penalty but separability, so the fitted row is a tie-break rather than an estimate [9, 12]. Intervene with the reproducible direction and the standard control, a norm-matched random vector, stays inert, while a rank-5 subspace fitted by the same estimator with the labels destroyed blocks transfer as strongly as the affect subspace itself.

The evidence is an estimator study across two models and three depths and three pre-registered experiments on the channel, each critiqued blind before its wording entered this paper. The response itself is robust to sampling: five of six emotions on each model. The channel is not what the linear picture suggests: removing the emotion direction everywhere it can be read leaves most of the response, only afraid and sad lose a fifth to a third, a rank-5 affect subspace cannot serve as an instrument because it shifts the readout used to check it, and a rank-1 permuted-label direction is inert while a rank-5 permuted-label frame is not.

The practical consequence is a change to the controls register of any ablation study of a concept direction: beside the random direction, run the same estimator on permuted labels at the rank of the intervention, report the footprint of every arm, and check the manipulation with an instrument that does not share the ablation’s target. The scientific consequence for affect transfer is narrower than we hoped: the transfer exists, the rank-1 direction carries part of it for two emotions, only sad’s part replicates across models, and what carries the rest is open.

Contributions.

- Dose-responsive movement of the receiver’s present-emotion projection for five of six emotions on two models, with common-random-number sampling and scenario-blocked inference; the receiver’s other-speaker projection moves comparably, which bounds the construct claim (Section 4).
- Disjoint-half reproducibility of probe directions as the missing quantity: logistic 0.57 / 0.29 against difference-of-means 0.97 / 0.94, unchanged by a fixed per-sample penalty, with the mechanism (perfect separation at every n ; the direction is the penalty’s tie-break and drifts from the mean difference) measured directly, and its downstream cost bounded by held-out accuracy and projection correlation (Section 3).
- Three pre-registered ablations with their verdicts as registered: rank-1 insufficiency (H1), rank-5 instrument failure, and survival of the rank-1 block against a permuted-label rank-1 control (H3), with replications of two designs on the second model (Section 5).
- Every arm against baseline in one table with one significance standard, including the two random controls the argument depends on, and the resulting rule for controls (Section 6).

Every number in this paper is registered with a pointer to the committed script and result file that produced it; the identifier appears in monospace after the number. Seven of our own interpretive sentences were withdrawn under blind critique, and the reviews of the first version forced eighteen more corrections; both lists are in the repository.

Relation to prior work. That logistic probe directions can be unstable is not our discovery. Marks and Tegmark [9] identified the deficiency and proposed mass-mean probing, noting that logistic regression converges toward the maximum-margin separator on separable data, the implicit bias analysed by Soudry et al. [12]; Gao et al. [5] name directional stability as a requirement for probe-then-steer pipelines and propose ridge-tuned logistic regression, reporting mean pairwise robustness

of 0.87–0.98 under 20% training-data ablations. Our contribution is the quantity: reproducibility across *disjoint* halves, which share no examples where two 80% subsamples share most, plateaus at 0.55–0.61 on the 27B and falls to 0.28–0.38 on the 8B while difference-of-means rises to 0.94–0.98 on the same splits. Braun et al. [2] corroborate for mean-difference steering vectors: refit on re-sampled draws of 5–30 activations they vary widely, and agree at cosine above 0.99 from 200–500 draws, which is the estimator we also find reproducible. Herbster et al. [6] report mean-difference and logistic directions as near-identical (0.98–0.99) for compassion; in a quantity close to theirs we find 0.43 and 0.33–0.37. Raju [11] measure stability of a representation across feature dimensions, a different axis.

2 Setup

Two agents, A and B, converse in 29 fixed scenarios. A’s message is generated under a steering vector added at layer 42 of Qwen3.6-27B (revision 6a9e13bd) at doses 0, 0.33, 0.67 and 1.0 of a fixed scale (the activation RMS at the focus layer), with three replicate seeds; B’s reply is then sampled with common random numbers across experimental arms, so that sampling noise shared between two arms cancels inside each paired contrast. Two readouts: B’s *present-emotion projection*, the projection of B’s reply activations on a difference-of-means probe for B’s own emotion, and B’s *other-speaker projection* on the matching probe for the interlocutor’s emotion; both probes are fitted on one half (808 items) of a 1,615-item probe pool while the steering and ablation directions are fitted, per layer, on the disjoint other half (807 items). A generation-side forced-choice readout of A’s span is the manipulation check. Every dose-response slope is an ordinary least-squares slope over the four doses on per-scenario means; every confidence interval is a scenario-blocked percentile bootstrap [4] (29 clusters, 5,000 draws), every between-arm contrast is scenario-blocked and dose-paired, and false-discovery rates are Benjamini–Hochberg [1] within one family of five testable emotions unless stated.

Two pools. The estimator study (Section 3) uses a separate, larger generation pool built for it: 4,253 items on the 27B (kept from 5,760 generated) and 3,898 on the 8B, so that disjoint halves of up to 2,000 items (27B) and 1,200 items (8B) exist. The ablation experiments use the 1,615-item probe pool above, on which the deployed directions’ own split-half cosine at 600 per half is 0.88–0.93 across the ablated layers (the pre-registered gate).

Footprint. For every ablation arm we record the norm of the component removed from A’s residual stream, averaged over the masked positions and the ablated layers; all ratios are relative to the emotion direction’s value in the same run (Table 3).

3 The estimator: reproducibility and its mechanism

Disjoint-half reproducibility. We fit the same estimator on two disjoint halves of n items each, ten independent splits, and report the mean per-class cosine between the two fitted directions (Figure 1). At the 27B focus layer with $n = 2000$ per half, difference-of-means reaches 0.974 and logistic regression ($C = 0.5$) 0.566 (a2.focus_dom_split_half, a2.focus_logreg_split_half); at layer 54, 0.979 against 0.554 (a2f54.*); on the 8B focus layer at $n = 1200$, 0.941 against 0.286 (a2_8b.*). Tuning C per fit by cross-validation reaches 0.61 on the 27B (a2f.focus_tuned_c_does_not_close_gap) and 0.38 at the 8B focus layer (0.37–0.41 across three depths; a2f8.1am_declines, note).

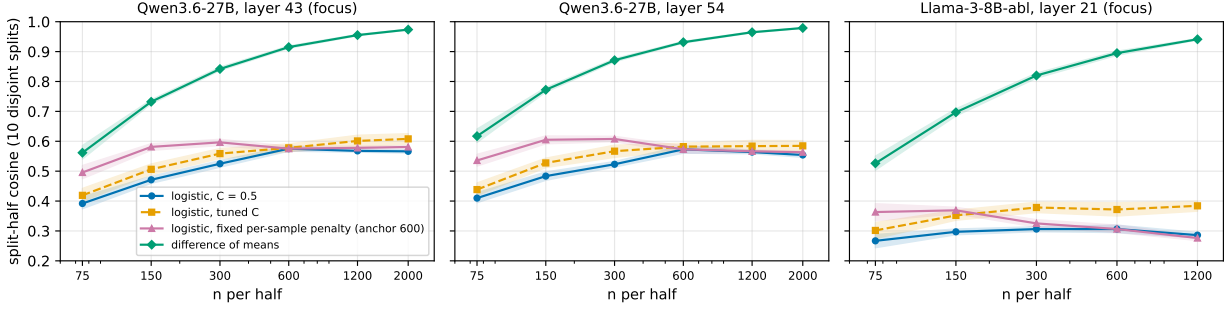


Figure 1: **Logistic directions do not reproduce; difference-of-means does.** Split-half cosine of the fitted direction over ten disjoint splits against n per half, at two depths of the 27B and the focus layer of the 8B. Difference-of-means rises toward 1 with n ; logistic at $C = 0.5$ and at a fixed per-sample penalty (anchor 600) plateau near 0.55–0.6 on the 27B and fall to 0.28–0.33 on the 8B; tuned C reaches 0.61 (27B) and 0.38 (8B). Bands are bootstrap CIs over splits.

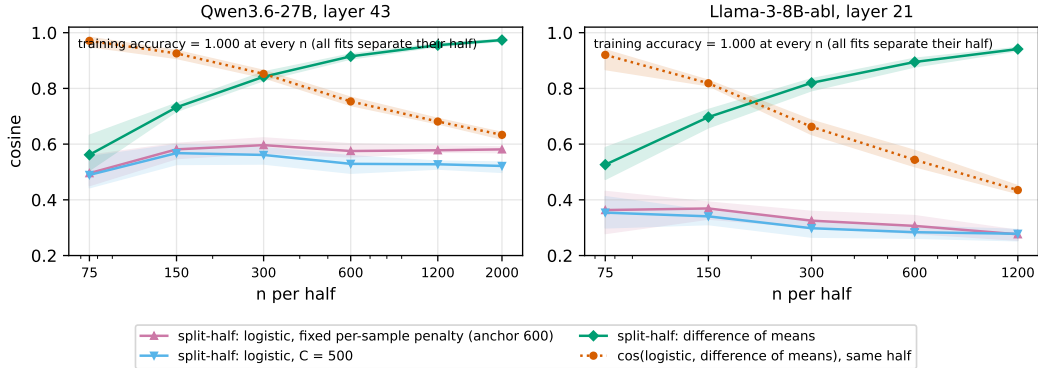


Figure 2: **The logistic direction is a separator’s tie-break that drifts away from the mean difference.** Within one half, the cosine between the fixed-penalty logistic direction and the difference-of-means direction falls as n grows, while difference-of-means becomes reproducible across halves and the logistic direction does not, under the fixed schedule (anchor 600) or a thousandfold weaker penalty ($C = 500$) alike. Every fit has training accuracy 1.000.

Holding the per-sample penalty fixed does not help. A natural objection is that a fixed C changes the effective penalty as n grows. With $C_n = 0.5 N_{\text{ref}}/n$, so that the per-sample penalty is constant, the logistic curve shows no net rise from $n = 150$ to 2000 at two anchors on the 27B focus layer (0.581 at both ends for the 600 anchor; a dip at 600 of -0.021 that is significant in a paired test over the ten shared splits, 1 of 10 positive, and a partial recovery; a2f.focus_logreg_fixed_lambda_flat, a2f2.lam_anchor2000_no_rise), one drop at 300→600 and then flat at layer 54 (a2f54.lam_declines_anchor600), a rise at layer 16, and on the 8B a peak at 150 followed by a fall to 0.276 at 1200 (a2f8.lam_declines). The picture is depth-specific, with no convergence over the reachable range ($n \leq 2000$, $d = 5120$ on the 27B, a 27-fold increase in n ; $n \leq 1200$, $d = 4096$ on the 8B): a proportional-regime statement, not an asymptotic one.

Why. Every multinomial logistic fit on every half at every n used, under the fixed schedule and $C = 0.5$, 50 and 500 alike (the two weak arms being one tolerance-limited solution), separates

its training half perfectly: accuracy exactly 1.000 in every cell, arm and seed; worst single-fit log-loss 0.0055; at most 80 iterations (`a2m.separable_every_n`, `a2m.8b_same_regime`, `a2m.v2_ten_seeds`). The likelihood is saturated from $n = 75$: the direction is the penalty’s tie-break among separating hyperplanes. Weakening the penalty a hundredfold or a thousandfold leaves the direction unchanged on the same half (cosine ≥ 0.96) and no more reproducible (0.52 against 0.58 at $n = 2000$; `a2m.weak_penalty_same_direction`). The separator starts near the mean-difference direction and departs from it as n grows, cosine $0.97 \rightarrow 0.63$ on the 27B (`a2m.logistic_departs_from_dom`) and $0.91 \rightarrow 0.43$ on the 8B (`a2m.8b_same_regime`), while difference-of-means becomes reproducible (Figure 2). In these units difference-of-means is the raw-space direction $(\mu_+ - \mu_-)/\sigma^2$; the plain mean difference $\mu_+ - \mu_-$ that contrastive activation addition uses agrees with the logistic direction at 0.43 (27B) and 0.33–0.37 (8B) (`a2f.caa_raw_vs_logreg_raw`, `a2f8.cross_estimator`), against the 0.98–0.99 that Herbster et al. [6] report for compassion. Not claimed: that the fit sits at the max-margin limit (the weak arms are tolerance-limited), or a support-vector mechanism for the drift.

Is this just $n < d$, and does it cost anything? Separation is guaranteed for generic data whenever $n < d$, so the mechanism is regime-conditional and the paper’s recommendation holds because probe pools in this literature are small relative to model width. Three checks bound what this means. On held-out items at the 27B focus layer the logistic probe decodes *better* than difference-of-means (six-way accuracy 0.82 to 0.94 against 0.79 to 0.88 at $n = 75$ to 2000), and the two estimators’ projections of the same held-out items correlate at 0.96 to 0.90 over the same range (`a21.heldout_and_projcorr`): the coefficient row is a fine classifier and a poor direction, the dissociation Marks and Tegmark [9] described, and the instability is largely geometric. Projecting the activations onto their top 100 principal components and refitting shows the regime dependence directly: with n/d' up to 20 the fits stop separating their halves from $n = 600$ (training accuracy 0.999, 0.989, 0.974 at $n = 600, 1200, 2000$), the logistic direction becomes reproducible (0.90 at $n = 2000$ in standardised coordinates against 0.94 for difference-of-means, from 0.39 and 0.33 at $n = 75$), and its cosine to the difference-of-means direction stays at 0.91–0.94 across n instead of falling from 0.90 to 0.62 as it does at full width; held-out accuracy in 100 dimensions is 0.92 against 0.94 at full width (`a21.lowd_recovers`). The 8B behaves the same way (`a21.8b_lowd`). The mechanism is therefore the $n < d$ regime itself, and the recommendation holds because probe pools in this literature are small relative to model width.

4 Transfer

B’s present-emotion projection rises with A’s dose for desperate, afraid, happy, sad and angry, with none-arm slopes from +48 to +212 and CIs excluding zero; calm’s (+44 [−3, +99]) does not, so calm is untestable under the pre-registered rule (`b1c.verdict_H1`; Figure 3). The slope is a poor summary for two emotions: happy rises +71 [+50, +94] over the first dose step and +0 [−24, +25] over the last, calm +38 [−3, +78] then −34 [−77, +13], while desperate, sad and angry rise at both steps (`rev.dose_shape`). The transfer is robust to sampling: the B1 follow-up run, with independently sampled replies, gives the same five testable emotions and afraid/sad as the blocked ones (`b1f.*`); B1c, B1d and B1e share the pool, the directions and the seed recipe and agree bit for bit on their shared arms, which is a determinism check, not a replication (`b1d.replicates_b1c_arms`). The manipulation check on A passes for all six emotions in each run. On the 8B the same five emotions respond (`b1c8.transfer_replicates`), with a steering manipulation half as strong and less specific, so only the pattern is compared.

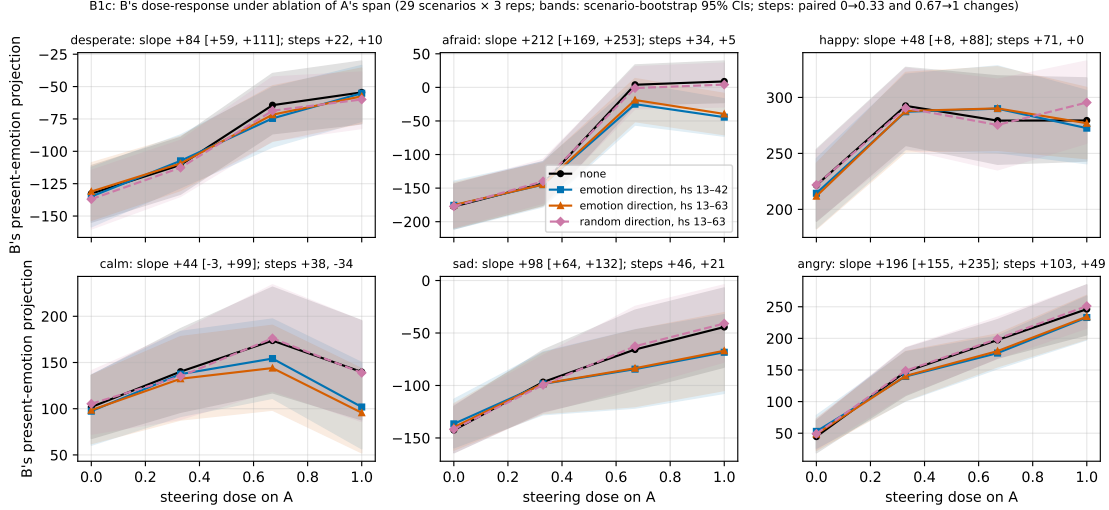


Figure 3: **The projection responds to steering; two of six responses are not monotone, and the ablation arms separate from baseline to significance for afraid and sad only among the testable emotions.** B’s present-emotion projection against A’s steering dose, per emotion, under no ablation and under the three ablation arms of B1c (29 scenarios $\times$ 3 replicates; bands are scenario-bootstrap 95% intervals of the per-dose mean). Titles give the none-arm slope with its CI and the paired changes over the first and last dose steps. Happy saturates after the first step and calm reverses at the top dose; calm’s separation is the largest in the figure but calm is untestable.

What the probe measures. B’s other-speaker projection, the matching probe for the interlocutor’s emotion, rises with A’s dose as well: slopes +59, +119, +65, +93, +96 for desperate, afraid, happy, sad and angry against present-probe slopes +84, +212, +48, +98, +196 (`rev.other_probe_rises`). The present-emotion projection therefore does not separate B’s own state from B’s representation of A’s state, and we make no claim about which it is; the earlier version of this work’s claim that the present shift exceeds the other shift did not survive a non-circular probe (report 18 in the repository).

5 Three pre-registered ablations

All directions are fitted per layer on the DIR half: the emotion direction at each hidden state 13–63 is that layer’s own difference-of-means direction; “readable” means the layer passes the pre-registered split-half gate (≥ 0.80), which every layer from 13 upward does and the window starts at $0.2\times$ depth by design.

B1c: the rank-1 direction (verdict H1). Projecting the difference-of-means emotion direction out of A’s span at every hidden state 13–63 during B’s prefill removes, against the random-direction control, a median 9% of the dose-response over the five testable emotions (scenario-bootstrap CI of the median [4%, 25%]; the per-emotion values are 0.06, 0.21, 0.05, 0.33, 0.09, and the median with calm included is 15%; `b1c.verdict_H1`, `b1c.median_fraction`, `rev.b1c_emo_vs_none_fdr`). Against baseline the median is 9% (17% with calm). Afraid (21%, $q = 0.001$) and sad (33%, $q = 0.048$) are the only blocks that clear the false-discovery rule against the random control; against baseline afraid clears

Table 1: Pre-registered decision contrasts, as blocked fractions of the none-arm slope (B1c: emotion direction – random direction; B1d: affect subspace – permuted-label subspace; B1e: emotion direction – permuted-label direction). *: BH-FDR $q < 0.05$ within the contrast’s own family of five testable emotions; ·: unadjusted CI excludes zero only. Calm (u) is untestable.

	desperate	afraid	happy	calm ^{<i>u</i>}	sad	angry
B1c: emotion dir – random dir	0.06	0.21*	0.05	(0.94)·	0.33*	0.09
B1d: affect subspace – permuted subspace	0.14	−0.13·	0.05	(1.56)·	0.26·	0.17·
B1e: emotion dir – permuted-label dir	0.06	0.25*	−0.14	(1.02)·	0.29*	0.09

Table 2: Every hooked arm against baseline (none), as blocked fractions of the none-arm slope, scenario-blocked and dose-paired; marks as in Table 1, families being the arm’s five testable emotions (`rev.b1c_random_vs_none`, `rev.b1c_emo_vs_none_fdr`, `rev.b1d_controls_vs_none`).

run	arm – none	desperate	afraid	happy	calm ^{<i>u</i>}	sad	angry
B1c	emotion direction (rank 1, hs 13–63)	+0.08	+0.24*	−0.24	(+0.98)·	+0.28·	+0.09
B1c	emotion direction (rank 1, hs 13–42)	+0.03	+0.27*	−0.11	(+0.79)·	+0.33*	+0.12*
B1c	random direction (rank 1)	+0.02	+0.03	−0.28	(+0.04)	−0.04	−0.00
B1d	affect subspace (rank 5)	+0.16	+0.11·	−0.05	(+1.32)·	+0.54*	+0.40*
B1d	permuted-label subspace (rank 5)	+0.03	+0.23*	−0.10	(−0.24)	+0.27*	+0.23*
B1d	random 5-frame (rank 5)	+0.05	+0.02	−0.17	(−0.12)	−0.01	+0.06
B1e	permuted-label direction (rank 1)	+0.02	−0.00	−0.10	(−0.04)	−0.01	+0.00
B1e	top principal direction (rank 1)	+0.16	−0.01	+0.30	(+0.98)·	−0.13	+0.61*

($q = 0.001$) and sad does not ($q = 0.115$) (`b1c.partial_block_replicates`). Extending the ablation to layers 43–63 changes no dose-response detectably, with the extra blocking bounded under 0.08 for afraid, angry, happy and sad (a driver-declared contrast; `b1c.no_gain_beyond_layer42`). The unit-norm random direction, removing twice the norm, is inert against baseline for all six emotions (largest happy +13.5 [−0.1, +28.4], $q \geq 0.26$; `rev.b1c_random_vs_none`). Across the five testable emotions the blocked fraction rises with the share of readable affect the ablation removes ($r \approx 0.85$ on five points, $p \approx 0.07$, descriptive; `b1c.readout_residual`): the rank-1 direction is insufficient to carry the transfer.

B1d: the rank-5 subspace (verdict instrument_failed). Projecting out the rank-5 class-mean affect subspace, which captures 16–21% of the pool’s variance and has cosine 0.71–0.81 to the top-5 principal components (`b1d.frame_geometry`), removes at least 80% of the readable affect for only three of six emotions and shifts the readout at dose 0 (desperate 0.11→0.50; `b1d.readout_baseline_shift`): the instrument and the ablation share a target, and the pre-registered decision is void. Against baseline (Table 2) the affect subspace blocks sad (54%) and angry (40%) at $q = 0.001$; a permuted-label subspace built by the same estimator, with no emotion information and a third of its span in common with the affect frame, blocks afraid (23%), sad (27%) and angry (23%) at $q = 0.001$, 0.035 and 0.013 (`b1d.perm_control_blocks`, `rev.b1d_controls_vs_none`), with dose-0 readout shifts below 0.08; a random 5-frame with half its footprint blocks nothing ($q \geq 0.51$). The pre-registered contrast, affect minus permuted subspace, clears no false-discovery threshold ($q = 0.051$; `b1d.median_fraction_descriptive`); the affect subspace differs from the rank-1 direction in opposite directions (afraid +29.3 slope units, i.e. less blocking; angry −60.6, more; `b1d.rank5_vs_rank1`).

Table 3: The directions and frames, one row each: rank, footprint (norm removed per masked position, averaged over ablated layers) with its ratio to the emotion direction in the same run, dose-0 shift of the A-span readout (largest over emotions), and share of readable affect removed at the top dose (range over emotions). (`b1c.random_control_inert`, `b1d.footprint`, `b1e.verdict_H3`, `b1e.pci_dissociation`, `rev.b1d_controls_vs_none`)

run	arm	rank	footprint (ratio)	dose-0 shift	readable affect removed
B1c	emotion direction	1	0.46–0.66 (1)	≤ 0.01	0.16–0.79
B1c	random direction	1	1.15–1.26 (2.0)	≤ 0.01	≤ 0.03
B1d	affect subspace	5	6.9 (10.5)	+0.39 / −0.25	0.11–1.25
B1d	permuted-label subspace	5	6.5–6.9 (10)	≤ 0.08	0.07–0.52
B1d	random 5-frame	5	3.1–3.2 (5)	≤ 0.01	≤ 0.03
B1e	permuted-label direction	1	0.50 (0.85; afraid 0.84, sad 0.77)	≤ 0.01	≤ 0.09
B1e	top principal direction	1	3.47 (5.8)	+0.55 / −0.28	0.71–1.5

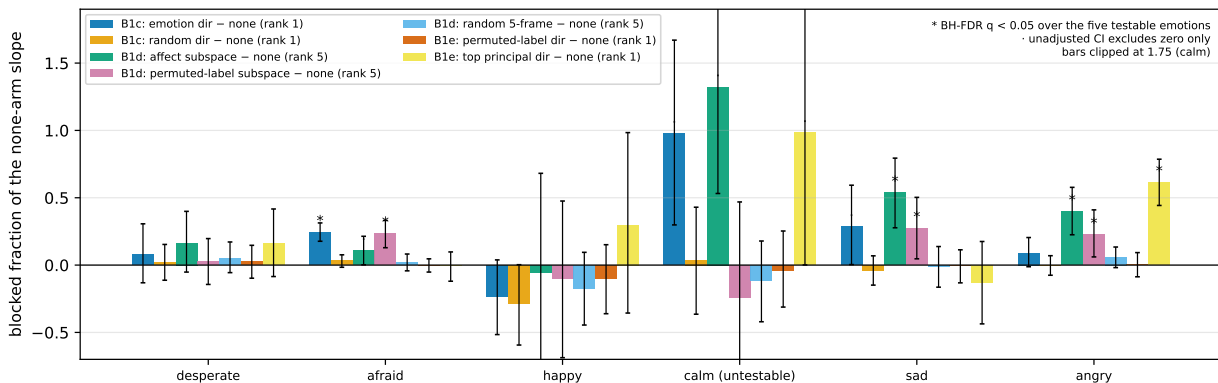


Figure 4: **Every arm against baseline: which ablations block, and which controls bite.** Blocked fraction of the none-arm slope for every hooked arm of the three 27B experiments (Table 2). The permuted-label rank-5 subspace blocks afraid, sad and angry at BH-FDR $q < 0.05$; the random direction, the random 5-frame and the permuted-label rank-1 direction block nothing; the top principal direction blocks angry. Error bars are CI-implied ranges; calm’s bars are clipped.

B1e: a permuted-label rank-1 control (verdict H3). The same estimator on permuted labels gives a rank-1 direction with cosine 0.004 to the emotion direction (measured before launch, not stamped in the file) and $0.85\times$ its footprint on average (0.84 afraid, 0.77 sad); removed at all 51 layers it changes no emotion’s dose-response and removes no readable affect (`b1e.permdir_inert`). Against it the emotion direction blocks afraid by 25% [17%, 33%] and sad by 29% [7%, 56%] (BH-FDR q 0.001 and 0.028; `b1e.verdict_H3`); the corresponding blocks against baseline are 24% and 28%. The top principal direction of A’s span, label-free but not fitted to labels, with $5.8\times$ the footprint, blocks angry by 61% ($q = 0.001$) and calm (knife-edge, untestable) while leaving afraid and sad untouched even though it removes 79% and 71% of their readable affect, and it shifts the dose-0 readout by up to 0.55 (`b1e.pci_dissociation`): readable-affect removal and blocking dissociate. B1d’s rank-5 permuted frame blocks where B1e’s rank-1 permuted direction does not; they differ in rank and in footprint ($10\times$ against $0.85\times$ the emotion direction’s), and which matters is untested: the band between the random 5-frame ($5\times$, inert) and the permuted frame ($10\times$, blocks) has no fitted control in it. The secondary readout on B’s reply agrees for sad and not for afraid (Appendix B).

Replication on the 8B. B1c’s design blocks the same two emotions on the 8B by unadjusted CIs (afraid 15%, sad 31% against the random direction; `b1c8.verdict_and_block`), with the other three too wide to tell agreement from disagreement; its manipulation-check rule returned `instrument_failed` because the upper ablation window changes the readout by nothing measurable on this model and the strict criterion has no tolerance band. B1e’s design returns H3 on the 8B with sad blocked by 36% (q 0.002, secondary readout agreeing) and afraid by 14% (q 0.08, no block against baseline); the control is matched only on average (sad $0.86\times$), both directions remove about the isotropic norm, and B1c’s random direction on that model reproduced the contrasts, so the run adds no discriminating power there. Only sad replicates across models on both outcomes; afraid is unreplicated (`b1e8.verdict_H3_matched`).

6 What the controls teach

Table 2 puts every arm on one scale. The random direction, the random 5-frame and the permuted-label rank-1 direction are inert; the permuted-label rank-5 frame blocks three emotions at the false-discovery threshold, as strongly as the affect subspace blocks two of them; the top principal direction blocks a different emotion. So a control that is fitted by the same estimator at the same rank as the intervention can bite where a random control of the same or larger footprint does not, and a study that reports only the random control would have called B1d’s blocks emotion-specific. Whether the fitted rank-5 frame bites because of its rank or its footprint is an open question that one further arm would settle (a random 5-frame scaled to the permuted frame’s footprint). Until then the rule is procedural: beside the random direction, run the same estimator on permuted labels at the rank of the intervention, report every arm’s footprint and dose-0 readout shift, and check the manipulation with an instrument that does not share the ablation’s target.

7 Limitations and what is not claimed

The present-emotion projection does not separate B’s own state from B’s representation of A’s; the other-speaker projection moves comparably. One probe pool and one direction estimate underlie all ablation runs on the 27B, and no ablation interval reflects the direction’s own sampling variability (the deployed directions’ split-half cosine is 0.88–0.93 at 600 per half). The A-span readout is forced-choice and shares a target with the subspace ablation. The dose-response slope summarises a saturating curve for happy and afraid and a reversing one for calm. All intervals are percentile bootstraps over 29 scenario clusters; multiplicity is controlled within experiments, not across the three experiments and two models. The estimator mechanism is regime-conditional ($n < d$): the low-dimensional check reproduces it and $n > d$ on the full width is not reached. Provenance limits and the defects recorded under review are listed in Appendix B; seeds, hyperparameters, hardware and software are in Appendix A. Not tested: the token channel (three rewrite designs failed their own manipulation check: the neutral rewrite kept 67–98% of the readable affect and the affect-preserving rewrite changed the transfer itself; `b1c.text_arms_uninterpretable`), the behavioural reach, layers below 13, and an ablation rerun with the logistic direction.

8 Conclusion

The idea to take away is not that affect transfers between agents, though something does, nor that logistic probes are unstable, which was known. It is that the two failures we found are the same failure seen twice. A logistic direction on separable activations is decided by the penalty’s tie-break,

so two independent fits agree only as far as their tie-breaks do; and a control direction is informative only if it is decided the way the intervention direction was. Random vectors are decided by nothing, which is why they are inert and why they miss what a fitted control catches. Any study that reads a concept with a probe and intervenes with its direction should report disjoint-half reproducibility of the direction, use an estimator that is an average rather than an extremum, and run its ablations against a control fitted by the same estimator on permuted labels at the intervention’s rank. For affect transfer specifically, the linear rank-1 direction carries a fifth to a third of the effect for two emotions and none of it for the others, one of the two replicates, the probe cannot yet say whose state it reads, and the next experiments are the two that would settle the open questions here: a footprint-scaled random frame, and an intervention that separates the receiver re-reading the sender’s tokens from the receiver reading the sender’s residual stream.

A Reproducibility statement

Code, pre-registrations, result files and the claim registry are in the accompanying repository (research-use licence). Models: Qwen3.6-27B revision **6a9e13bd** and Llama-3-8B-Instruct-abliterated revision **dd67dd05**, run with torch 2.11.0+cu130 and transformers 5.14.1 on one NVIDIA RTX PRO 6000 (96 GB) per run; a 27B ablation experiment takes about two hours (432 or 288 arm-rows of 29 scenarios each, 19–21 s per arm-row), the 8B one under thirty minutes. Seeds: pool generation seed 0; per-generation seeds are a CRC32 of (seed, emotion, dose, replicate, arm, part) with the arm field empty for the receiver’s reply so that arms share random numbers; ten split seeds (0–9) for every split-half number; bootstrap seed 0 with 5,000 draws. Hyperparameters: doses $\{0, 0.33, 0.67, 1\}$ at a fixed activation-RMS scale, three replicates, 29 scenarios, 110 new tokens per message at temperature 0.9; logistic regression with lbfgs, max_iter 3000, $C \in \{0.5, C_n = 0.5 N_{\text{ref}}/n \text{ with } N_{\text{ref}} \in \{600, 2000\}, 50, 500\}$ and cross-validated C on a decade grid scored for accuracy; split-half $n \in \{75, 150, 300, 600, 1200, 2000\}$ per half where the pool allows. Statistics: scenario-blocked percentile bootstraps because scenarios are the sampling unit; between-arm contrasts paired within scenario and dose; two-sided bootstrap p -values with Benjamini–Hochberg over the testable emotions of one arm; every interval in the paper is such a bootstrap and every count of runs is stated. The probe pools and scenario lists are committed with the results.

B Details and defects recorded under review

Each item below is in the registry note of the claim it qualifies.

- *B1c*. The contrast between the 30-layer and 51-layer windows was declared in the driver, not in the pre-registration; its null is unpowered for desperate and calm. The testability rule (none-arm CI excludes zero) is pre-registered (PREREG_B1c §5). Calm’s contrast is the largest and is excluded by that rule; medians are reported with and without it. Afraid’s top-dose readout is non-monotone. The pre-registration stamp is empty because the driver resolved its path relative to the wrong directory; the document was on the machine one second before the run started, and its commit (**cc2dc32**, 19:22Z) precedes the run’s start (19:32Z) in the repository history. The pool has 1,615 items against the registered 2,160; the false-discovery rate ran over five emotions rather than six.
- *B1d*. The permuted-label subspace removed only 6.5% of afraid’s readable affect and still blocked its transfer, so overlap with the readable-affect direction does not explain its blocking;

recomputing the instrument share with each arm’s own floor keeps three failures but changes which emotions fail. The median’s confidence interval resamples scenarios only.

- *B1e*. The control removes less norm than the emotion direction and less than B1c’s inert random direction (1.22), so it is a fitted label-free null, not a footprint-matched one. The rule tests the point fraction and sad’s range starts at 0.07. The secondary readout on B’s reply contradicts afraid’s block and agrees for sad. The control’s cosine to the emotion direction (0.004) was measured by the pre-launch verifier and is not stamped in the result file.
- *8B replications*. Both are out-of-pre-registration replications (the documents name the 27B and its layers). The steering manipulation is half as strong and less specific on the 8B, afraid’s readout sits below chance, and slopes are unnormalised probe scores, so only patterns are compared. In B1e on the 8B the control is matched on average only (sad $0.86\times$), both directions remove about the isotropic norm, and B1c’s random direction already reproduced the contrasts; afraid does not block against baseline there.
- *Estimator study*. The weak-penalty arms ($C = 50$ and 500) give the same solution to three decimals, so they are tolerance-limited and bound nothing about the $C \rightarrow \infty$ limit. Cross-estimator cosines use subsamples that overlap by about half the pool, so their intervals overstate precision. The layer-43 and layer-54 files come from three driver builds; the shared cells agree bit for bit.

References

- [1] Yoav Benjamini and Yosef Hochberg. Controlling the false discovery rate: A practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society: Series B*, 57 (1):289–300, 1995.
- [2] Joschka Braun, Carsten Eickhoff, David Krueger, Seyed Ali Bahrainian, and Dmitrii Krashennnikov. Understanding (Un)Reliability of steering vectors in language models. *arXiv preprint arXiv:2505.22637*, 2025.
- [3] Funda Durupinar. How affect propagates among LLM agents: Emergent emotional contagion in crowd simulation. *arXiv preprint arXiv:2607.25140*, 2026.
- [4] Bradley Efron and Robert J. Tibshirani. *An Introduction to the Bootstrap*. Chapman and Hall, 1993.
- [5] Ziqi Gao, Yaotian Zhu, Qingcheng Zeng, Xu Zhao, Ziqing Wang, Feng Ruan, and Kaize Ding. RAPTOR: Ridge-adaptive logistic probes. *arXiv preprint arXiv:2602.00158*, 2026.
- [6] Niklas Herbster, Martin Zborowski, Alberto Tosato, Gauthier Gidel, and Tommaso Tosato. Activation steering for aligned open-ended generation without sacrificing coherence. *arXiv preprint arXiv:2604.08169*, 2026.
- [7] Kenneth Li, Oam Patel, Fernanda Viégas, Hanspeter Pfister, and Martin Wattenberg. Inference-time intervention: Eliciting truthful answers from a language model. *arXiv preprint arXiv:2306.03341*, 2023.
- [8] Yunbo Long, Liming Xu, Lukas Beckenbauer, Yuhan Liu, and Alexandra Brintrup. EvoEmo: Towards evolved emotional policies for adversarial LLM agents in multi-turn price negotiation. *arXiv preprint arXiv:2509.04310*, 2025.

- [9] Samuel Marks and Max Tegmark. The geometry of truth: Emergent linear structure in large language model representations of true/false datasets. *arXiv preprint arXiv:2310.06824*, 2023.
- [10] Nina Panickssery, Nick Gabrieli, Julian Schulz, Meg Tong, Evan Hubinger, and Alexander Matt Turner. Steering Llama 2 via contrastive activation addition. *arXiv preprint arXiv:2312.06681*, 2023.
- [11] Prashant C. Raju. Geometric stability: The missing axis of representations. *arXiv preprint arXiv:2601.09173*, 2026.
- [12] Daniel Soudry, Elad Hoffer, Mor Shpigel Nacson, Suriya Gunasekar, and Nathan Srebro. The implicit bias of gradient descent on separable data. *Journal of Machine Learning Research*, 19(70):1–57, 2018. arXiv:1710.10345.
- [13] Alexander Matt Turner, Lisa Thiergart, Gavin Leech, David Udell, Juan J. Vazquez, Ulisse Mini, and Monte MacDiarmid. Steering language models with activation engineering. *arXiv preprint arXiv:2308.10248*, 2023.
- [14] Andy Zou, Long Phan, Sarah Chen, James Campbell, Phillip Guo, Richard Ren, Alexander Pan, Xuwang Yin, Mantas Mazeika, Ann-Kathrin Dombrowski, et al. Representation engineering: A top-down approach to AI transparency. *arXiv preprint arXiv:2310.01405*, 2023.